

Stability and Station-Keeping Near the Earth-Moon L_2 in Planar CRTBP

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I. INTRODUCTION

A restricted three-body problem (RTBP) depicts a system where two major celestial bodies is orbited by a third body of negligible mass. For most solar system applications, it can be further assume that mass of the smaller body is negligible compared to the bigger one. This forms the circular restricted three-body problem (CRTBP). CRTBP is commonly used as a nonlinear dynamical model for deep space missions as a convenient substitute for more complicated n-body integrators. Lagrange points are a set of five equilibria in CRTBP. They are formed at the balance between the gravitational and centrifugal forces in a rotating frame, thus their positions in such a frame rely only on the mass and distance of the two bodies [1].

The second Lagrange point, denoted as L_2 , is of particular interest to modern space missions, as it is surrounded by a family of periodic trajectories convenient for maintaining a spacecraft's position. For example, the James Webb Space Telescope is positioned at Earth-Sun L_2 to reduce the pressure on its cryogenic cooling using Earth's shadow [2]. Similarly, the Chinese Queqiao satellite, which orbits the Earth-Moon L_2 to provide constant communication coverage for Chang'e missions on the far side of the Moon [3]. This family of periodic Lagrange point-orbiting trajectories is known as the Lyapunov orbits. Based on whether the orbit lies entirely in the orbital plane of the smaller celestial object or not, Lyapunov orbits can be further classified as planar (horizontal) and vertical Lyapunov orbits, respectively. However, as this work will show later, L_2 is an unstable equilibrium because it corresponds to a saddle point of the effective potential in the rotating frame. Therefore, spacecraft utilizing orbits in this region will require closed-loop control.

This paper aims to analyze the stability of a spacecraft on a planar Lyapunov orbit in the Earth-Moon L_2 region, first uncontrolled, then under a closed-loop stabilizer.

II. PHYSICS PRELIMINARIES

Consider a reference frame that rotates synchronously with the Moon's motion around Earth. Set its origin at the Earth-Moon barycenter, and fix the Moon's barycenter along the positive x -axis. The positive z -axis of this reference frame is defined by the direction of the Moon's orbital angular momentum, and let positive y -axis being perpendicular with x and z , such that $\hat{y} = \hat{z} \times \hat{x}$, where $\hat{\cdot}$ denotes a unit

vector. This is conventionally referred to as the "synodic" frame. A spacecraft in this rotating synodic frame not only experiences the same forces as it would in a non-rotating frame (i.e. gravitational forces from both Earth and the Moon), it would also experience a fictitious centrifugal force. Since gravitational forces and centrifugal force are conservative, the total mechanical energy E of a spacecraft in the Earth-Moon System (EMS) is given by:

$$E = \frac{1}{2}mv^2 - \frac{1}{2}m\Omega^2 d^2 - \left(\frac{m\mu_{Earth}}{r_{Earth}} + \frac{m\mu_{Moon}}{r_{Moon}} \right) \quad (1)$$

$$\epsilon = v^2 - \Omega^2 d^2 - 2 \left(\frac{\mu_{Earth}}{r_{Earth}} + \frac{\mu_{Moon}}{r_{Moon}} \right) \quad (2)$$

where m is the object's mass, ϵ is its characteristic energy, v is the object's speed relative to the reference frame, Ω is the reference frame's rate of rotation in angle per second, and d is the Earth-Moon distance. $\mu = GM$ is the gravitational constant for the celestial body in subscript, and r is the spacecraft's distance to that body. In CRTBP literature such as [4], ϵ is denoted as C_J , the Jacobi Constant. Setting v in Equation 2 to zero results in a level set known as the zero-velocity curve (or "surface" for the three-dimensional cases), which defines the classically forbidden region and bounds the possible positions of the object. This expression will be used later to analyze the Lyapunov stability of the object.

Next, the equation of motion will be derived for a spacecraft near the Earth-Moon L_2 . The equation of motion will be used to construct a numerical simulation to verify the results of Lyapunov stability analysis. Since Lyapunov stability is scale-invariant, this part of analysis will be carried out using non-dimensional parameters.

Consider the Lagrangian L in a rotating frame, and throw out the z component since this work focuses only on the planar motions. The same convention as in [4] are adopted here, where total mass of the EMS, rate of rotation of the reference frame, as well as the separation between the two celestial bodies, are normalized to 1. The mass ratio of the smaller body is also set to be μ , then a dimensionless expression for L is obtained:

$$L(x, y, \dot{x}, \dot{y}) = \frac{1}{2}((\dot{x} - y)^2 + (\dot{y} + x)^2) - U(x, y) \quad (3)$$

where

$$U(x, y) = \frac{1 - \mu}{r_{Earth}} + \frac{\mu}{r_{Moon}} + \frac{\mu(1 - \mu)}{2} \quad (4)$$

Given the Euler-Lagrange equation,

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0 \quad (5)$$

Plug Equation 3 into it, and solve for the second derivatives to obtain the equation of motion:

$$\begin{cases} \ddot{x} = 2\dot{y} + \bar{U}_x \\ \ddot{y} = -2\dot{x} + \bar{U}_y \end{cases} \quad (6)$$

where $\bar{U}(x, y)$ is the dimensionless effective potential, and

$$\bar{U}(x, y) = \frac{1}{2}(x^2 + y^2) + \frac{1 - \mu}{r_{Earth}} + \frac{\mu}{r_{Moon}} + \frac{\mu(1 - \mu)}{2} \quad (7)$$

$\bar{U}_x$ and $\bar{U}_y$ are x and y partial derivatives of $\bar{U}$, respectively. At equilibrium, the position and velocity states are constant. In other words, at (x^*, y^*) , $\ddot{x}^* = \ddot{y}^* = \dot{x}^* = \dot{y}^* = 0$. Substituting these conditions into the equation of motion:

$$\begin{cases} \bar{U}_x = 0 \\ \bar{U}_y = 0 \end{cases} \quad (8)$$

Therefore, the equilibrium points must also be the critical points of effective potential in equation 7. To derive the r_{Earth} and r_{Moon} expressions in terms of x , y , and μ , recall that the sum of the gravitational constants has been set to 1, same for the Earth-Moon separation. At EMS barycenter, the mass-weighted average position of the two bodies is zero, thus:

$$(1 - \mu)x_{Earth} + \mu x_{Moon} = 0$$

which, given $x_{Moon} - x_{Earth} = 1$, implies that

$$\begin{cases} x_{Earth} = -\mu \\ x_{Moon} = 1 - \mu \end{cases} \quad (9)$$

are the distances of Earth and Moon from the EMS barycenter. For a spacecraft at (x, y) ,

$$\begin{cases} r_E = r_{Earth} = \sqrt{(x + \mu)^2 + y^2} \\ r_M = r_{Moon} = \sqrt{(x - 1 + \mu)^2 + y^2} \end{cases} \quad (10)$$

Plug these back into the effective potential and differentiate with respect to x and y :

$$\begin{cases} \bar{U}_x = x - (1 - \mu) \frac{x + \mu}{r_E^3} - \mu \frac{x - 1 + \mu}{r_M^3} \\ \bar{U}_y = y - (1 - \mu) \frac{y}{r_E^3} - \mu \frac{y}{r_M^3} \end{cases} \quad (11)$$

This appears intimidating at the first glance. Indeed, both the r_{Earth} and r_{Moon} terms contain both x and y , thus both equations are coupled with each other. However, for a spacecraft whose trajectory stays within a sufficiently small area containing L_2 , its distances to Earth and Moon remains

overall unchanged. Therefore, for stability analysis, r_{Earth} and r_{Moon} can be treated as constants. Combine Equations 8, 10, and 11, and set $y = 0$ to find the x position of L_2 :

$$x - \frac{(1 - \mu)}{(x + \mu)^2} - \frac{\mu}{(x - 1 + \mu)^2} = 0 \quad (12)$$

Recall that L_2 is the Lagrange point on the far side of the Moon, and recall that at equilibrium, $\ddot{x}^* = \dot{x}^* = 0$. Then L_2 would have a coordinate taking the form $(x^*, 0)$, with $x^* > 1 - \mu$. Solve Equation 12 numerically to find a root satisfying these conditions, with $\mu \approx 0.01215$ given in [4], the non-dimensional coordinate for L_2 is $(1.156, 0)$. This result roughly corresponds to the real-life location of Earth-Moon L_2 [5]. Therefore, this shows that L_2 is an equilibrium point indeed.

III. STABILITY ANALYSIS WITH THE UNCONTROLLED CASE

To set up the system for analysis, define the spacecraft's state as the combination of its position and velocity states, and use the equation of motion in Equation 6 to write the system as a system of ordinary differential equations:

$$X = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \end{bmatrix}, \dot{X} = f(X) = \begin{bmatrix} \dot{x} \\ \dot{y} \\ 2\dot{y} + \bar{U}_x \\ -2\dot{x} + \bar{U}_y \end{bmatrix} \quad (13)$$

Mission planning and orbital mechanics texts commonly refer to X as the "state vector". From examining Equation 6, the equation of motion for a spacecraft near L_2 is at least twice differentiable. Therefore, it would be possible to set up the Taylor expansion of $f(X)$, and take only the first order term for linearization. A first-order Taylor expansion gives

$$\delta \dot{X} = A \delta X \quad (14)$$

where

$$\delta X = X - X^* = \begin{bmatrix} \delta x \\ \delta y \\ \delta \dot{x} \\ \delta \dot{y} \end{bmatrix}, \text{ and } A = \left. \frac{\partial f}{\partial X} \right|_{X^*} \quad (15)$$

Note that A is a Jacobian whose each index is filled by a partial derivative $\frac{\partial f_i}{\partial q_j}$. Each f_i is a component of δX , and q_j a component of the state vector, for i and j denoting rows and columns of A , respectively. Compute each partial derivative and put them in the Jacobian:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \bar{U}_{xx} & \bar{U}_{xy} & 0 & 2 \\ \bar{U}_{xy} & \bar{U}_{yy} & -2 & 0 \end{bmatrix} \quad (16)$$

where

$$\begin{cases} \bar{U}_{xx} = 1 - (1 - \mu)\left(\frac{1}{r_E^3} - \frac{3(x+\mu)^2}{r_E^5}\right) - \mu\left(\frac{1}{r_M^3} - \frac{3(x-1+\mu)^2}{r_M^5}\right) \\ \bar{U}_{yy} = 1 - (1 - \mu)\left(\frac{1}{r_E^3} - \frac{3y^2}{r_E^5}\right) - \mu\left(\frac{1}{r_M^3} - \frac{3y^2}{r_M^5}\right) \\ \bar{U}_{xy} = 3(1 - \mu)\frac{(x+\mu)y}{r_E^5} + 3\mu\frac{(x-1+\mu)y}{r_M^5} \end{cases} \quad (17)$$

Since $L_2 = (x^*, 0)$, $\bar{U}_{xy} = 0$, $\bar{U}_{yy} = 1 - \frac{1-\mu}{r_E^3} - \frac{\mu}{r_M^3}$, the Jacobian at L_2 is evaluated to be:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \bar{U}_{xx} & 0 & 0 & 2 \\ 0 & \bar{U}_{yy} & -2 & 0 \end{bmatrix} \quad (18)$$

and the characteristic equation is

$$\lambda^4 + (4 - \bar{U}_{xx} - \bar{U}_{yy})\lambda^2 + \bar{U}_{xx}\bar{U}_{yy} = 0 \quad (19)$$

Evaluate using the μ and x^* values determined at the end of the previous section, four eigenvalues are found: $\lambda_{1,2} = \pm 2.1587$, $\lambda_{3,4} = \pm 1.8626i$. From the existence of an eigenvalue with positive real part, it can be immediately concluded that the system is unstable at L_2 , which is consistent with the current consensus. On the other hand, the other eigenvalue that has negative real part indicates that L_2 also has a stable manifold symmetric and opposite to the unstable one, and the imaginary pair indicates that at the intersection between the stable and unstable manifolds, a periodic trajectory may exist. This type of periodic orbits around a Lagrange point are known as Lyapunov orbits in CRTBP mission design texts [6].

Compute the eigenvectors corresponding to each of the eigenvalues above, and find:

$$\begin{aligned} v_s &= \begin{bmatrix} 0.356 \\ 0.224 \\ -0.768 \\ -0.484 \end{bmatrix}, v_u = \begin{bmatrix} 0.356 \\ -0.224 \\ 0.768 \\ -0.484 \end{bmatrix}, \\ v_{p1} &= \begin{bmatrix} 0.154 \\ 0 \\ 0 \\ -0.833 \end{bmatrix}, v_{p2} = \begin{bmatrix} 0 \\ 0.447 \\ 0.286 \\ 0 \end{bmatrix} \end{aligned} \quad (20)$$

where v_s corresponds to the stable mode, v_u the unstable mode, and v_{p1} and v_{p2} correspond to the periodic modes. Note that v_s has the same elements as v_u , but with signs reversed. This confirms the previous observation that a stable/unstable manifold pair exists around L_2 . Indeed, if a spacecraft receives perturbation, its position states will change more drastically along the $(0.356, -0.224)$ vector, and less so along the $(0.356, 0.224)$ vector. Furthermore, the eigenvectors associated with the periodic modes show that the periodic modes favor the type of trajectory that makes an elliptical shape in the x - y plane in the counter-clockwise direction. A simulated example in the next section will illustrate these observations.

IV. SIMULATION WITH THE UNCONTROLLED CASE

To verify the analytical results discussed in the previous section, three initial conditions are selected and fed into the equations of motion. Then, the equations of motion and initial conditions are integrated using a Python implementation of `DOP853`, which is an explicit 8th order Runge-Kutta method, and one that has the highest order among the methods offered by `scipy.integrate.solve_ivp` [7]. This solver is then left to run for 10 non-dimensional time units (equivalent to ≈ 43.5 days), or when the spacecraft completes one Lyapunov orbit at a position other than L_2 . The initial conditions and simulation outcomes are listed in Table I. The trajectories are plotted in Figure 1.

TABLE I
INITIAL CONDITIONS AND OUTCOMES FOR ALL THREE CASES, WITHOUT CONTROL INPUTS.

Cases	Initial condition	Outcome
1	At L_2	Stable at L_2
2	$x^* + 10^{-12}$	Spiraled out
3	$X_0 = [1.14568217, 0, 0, 5.253542e - 2]^T$	Periodic

First, the location of L_2 was recalculated using the more accurate data provided by [8], and it was used as the initial condition to generate case 1. Case 2 is meant to demonstrate the instability of L_2 , so a small perturbation of $\delta x = 10^{-12}$ was applied to the initial condition of case 1. The generation of case 3 takes advantage of the symmetry property of a planar Lyapunov orbit, and sets $\dot{x}_0 = 0$ at $y_0 = 0$. A manually adjustable offset was applied to x_0 to control the size and shape of the planar Lyapunov orbit, thus all that is left to vary is just the $\dot{y}_0$ parameter.

A shooting method is used to find the appropriate $\dot{y}_0$ with these initial conditions. Given the symmetry property above, the orbit must cross the x -axis twice per period, and have $\dot{x}(t) = 0$ at each crossing. Define a half-period event $t = t_{\frac{1}{2}}$ such that the trajectory crosses the x -axis for the first time after start. Define a tolerance function $F(\dot{y}_0) = \dot{x}(t_{\frac{1}{2}}, \dot{y}_0)$, and iterate through different values of $\dot{y}_0$ until the tolerance reaches sufficiently close to 0. Let $\bar{\dot{y}}_0$ be the value of $\dot{y}_0$ when $F(\dot{y}_0) = 0$.

Next, design an iteration strategy to minimize $F(\dot{y}_0)$. From a few rounds of trial-and-error with $\dot{y}_0 \approx \bar{\dot{y}}_0$, it can be observed that $F(\dot{y}_0) > 0$ when $\dot{y}_0 > \bar{\dot{y}}_0$, and vice versa. Informed by this observation, the neighborhood of the root for $F(\dot{y}_0)$ can be located manually via bisection, then computed to a sufficiently tight tolerance using gradient descent. The resulting planar Lyapunov orbit has tolerance $F(\dot{y}_0) < 1e - 12$, and its initial conditions are listed in Table I.

Even after being integrated for three orbits, case 3 still shows good periodicity, as illustrated in the clean, non-divergent orbit in Fig. 1 and its overlap with case 1 in the Poincaré map in Fig. 2 in $(x-\dot{x})$ phase space. Due to numerical errors, however, case 3's periodicity eventually breaks down after about five orbits.

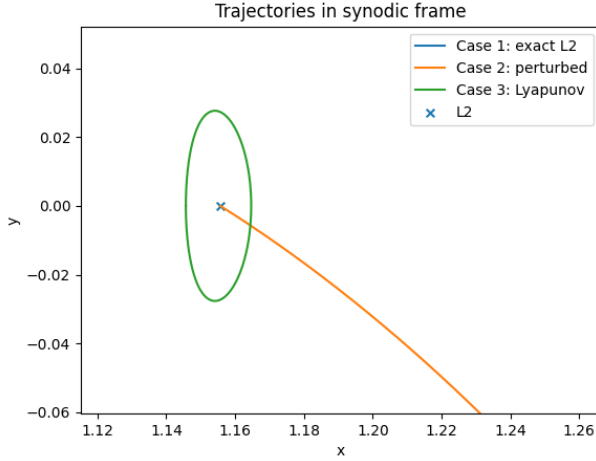


Fig. 1. Trajectories in a synodic frame, zoomed in to emphasize cases 1 and 3. The orange line for case 2 rapidly diverges into the bottom-right due to its instability.

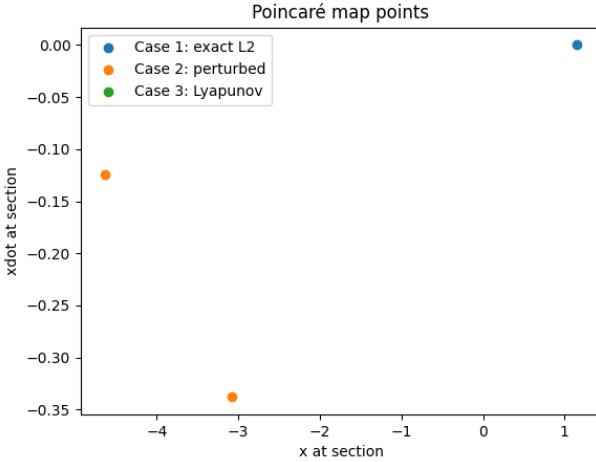


Fig. 2. Poincaré map of all three trajectories in $(x, \dot{x})$ phase space. The points for case 3 is overlapped with case 1 at this scale.

V. STABILITY ANALYSIS WITH LINEAR QUADRATIC REGULATOR

Up to this point, it has been shown with both analysis and simulation that L_2 is unstable, and although a family of periodic trajectories exist around it, these trajectories can easily lose their periodicity given a small perturbation. If a spacecraft has thrust capabilities in the x - y plane, controllability theorems indicate that there exists a control gain matrix K , such that for control input $u = -Kx$, every eigenvalue of the system $\dot{x} = Ax + Bu$ lies within $\mathbb{C}^-$.

Consider the equation of motion with control input in x and y :

$$\begin{cases} \ddot{x} = 2\dot{y} + \bar{U}_x(x, y) + u_x \\ \ddot{y} = -2\dot{x} + \bar{U}_y(x, y) + u_y \end{cases}, u = \begin{bmatrix} u_x \\ u_y \end{bmatrix} \quad (21)$$

The steps to linearize the system in equation 21 are essentially the same steps as equations 13 through 15, but with an added control term. Recall that

$$\delta \dot{X} = X - X^*, A = \left. \frac{\partial f}{\partial X} \right|_{X^*} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \bar{U}_{xx} & 0 & 0 & 2 \\ 0 & \bar{U}_{yy} & -2 & 0 \end{bmatrix}$$

With the assumption that the spacecraft is fully controllable using thrusters, which can only control the velocity states, the control matrix B is set up as such:

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (22)$$

together, the linearized error dynamics is characterized by the following equation:

$$\delta \dot{X} = A\delta X + Bu = (A - BK)\delta X \quad (23)$$

Next, design a stabilizer K using Linear Quadratic Regulator (LQR). LQR finds $K = R^{-1}B^TP$ by finding an appropriate P via solving the Algebraic Riccati Equation (ARE):

$$A^TP + PA - PBR^{-1}B^TP + Q = 0 \quad (24)$$

Start by setting $R = I_2$ and $Q = I_4$, then the ARE simplifies to:

$$A^TP + PA - PBB^TP + I = 0 \quad (25)$$

Plug in the previously determined x^* and μ values to find $\bar{U}_{xx}$ and $\bar{U}_{yy}$ for A , and yield:

$$P \approx \begin{bmatrix} 39.484 & -6.917 & 12.105 & 5.758 \\ -6.917 & 3.856 & -1.861 & -0.663 \\ 12.105 & -1.861 & 4.177 & 1.430 \\ 5.758 & -0.663 & 1.430 & 1.830 \end{bmatrix} \quad (26)$$

which gives

$$K \approx \begin{bmatrix} 12.105 & -1.861 & 4.177 & 1.430 \\ 5.758 & -0.663 & 1.430 & 1.830 \end{bmatrix} \quad (27)$$

The following eigenvalues are calculated from $(A - BK)$:

$$\begin{cases} \lambda_1 = -2.357 \\ \lambda_2 = -1.936 \\ \lambda_{3,4} = -0.857 \pm 1.788i \end{cases} \quad (28)$$

all of which reside within $\mathbb{C}^-$. The fact that all eigenvalues contain negative real parts indicates that the system is asymptotically stable.

To verify the system's Lyapunov stability using Lyapunov Direct Method, I picked the LQR value function as the Lyapunov function, which is given by:

$$V(\delta X) = \delta X^T P \delta X \quad (29)$$

where P is extracted from Equation 24. Since LQR requires P to be positive definite, which can be verified by checking that all the eigenvalues in Equation 26 are positive. Therefore, $V(\delta X)$ is also positive definite. Differentiate $V(\delta X)$ with respect to δX :

$$\begin{aligned} \dot{V} &= \delta \dot{X}^T P \delta X + \delta X^T P \delta \dot{X} \\ &= \delta X^T (A^T P + P A - K^T B^T P - P B K) \delta X \end{aligned} \quad (30)$$

Using the LQR identity that $K = R^{-1} B^T P$, which implies that $K^T B^T P = K^T R K$ and $P B K = (B^T P)^T K = K^T R K$, then:

$$\dot{V} = \delta X^T (A^T P + P A - 2K^T R K) \delta X \quad (31)$$

and ARE can be written as :

$$A^T P + P A = -Q + K^T R K \quad (32)$$

Combine the two equation above, and

$$\dot{V} = -\delta X^T (Q + K^T R K) \delta X \quad (33)$$

Since Q and $K^T R K$ are both positive definite given the choice of Q and R , $\dot{V}$ must be negative definite at X^* . Therefore, L_2 has turned into an asymptotically stable equilibrium given the LQR controller.

VI. SIMULATION WITH THE CONTROLLED CASE

Recall that $u = -K\delta X = -K(X - X^*)$, then

$$\begin{cases} u_x = -k_1[(x - x^*), y, \dot{x}, \dot{y}]^T \\ u_y = -k_2[(x - x^*), y, \dot{x}, \dot{y}]^T \end{cases}, K = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} \quad (34)$$

Combining Equation 34 with Equation 21 gives the full equation of motion that incorporates the controller. Since the resulting equation takes on the same form as Equation 6, the same Python script used in Section IV can be reused with only modification to the equations of motion. Since the eigenvalues with closed loop control all have strictly negative real parts, the periodic case is no longer necessary for demonstrating stability about L_2 . Hence, the initial conditions are re-selected to represent various levels of perturbation. These initial conditions and their outcomes are listed in Table II.

TABLE II
INITIAL CONDITIONS AND OUTCOMES FOR ALL THREE CASES, WITH CONTROL INPUTS PRODUCED BY LQR.

Cases	Initial condition	Outcome
1	At L_2	Stable at L_2
2	$x^* + 10^{-12}$	Stable at L_2
3	$\delta X_0 = [10^{-3} \quad 10^{-3} \quad 10^{-3} \quad -10^{-3}]^T$	Converges to L_2

Again, use `scipy.integrate.solve_ivp` with DOP853 to integrate the equation of motion to produce Fig. 3 and Fig. 5. Both figures clearly show that the LQR controller has converted an unstable equilibrium into a stable one, confirming the earlier analysis. Interestingly, the convergence trajectory for case 3 follows the counter-clockwise ellipse-like path, which was previously theorized for the uncontrolled case in Section III.

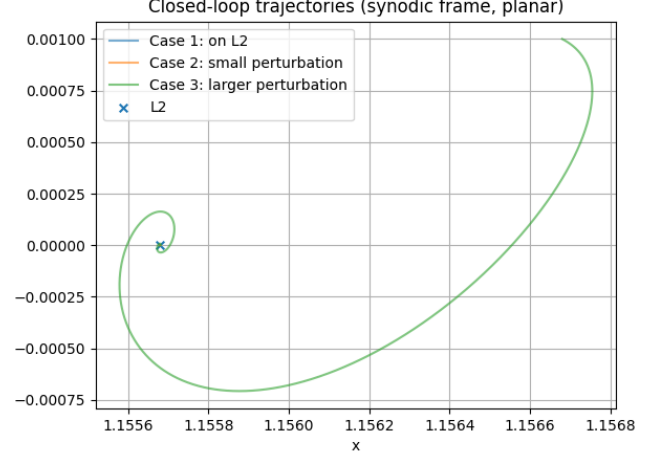


Fig. 3. Trajectories of all three cases in (x, y) space, with LQR. $R = I_2$ and $Q = I_4$. Note that case 3 (green trajectory) starts from the top right.

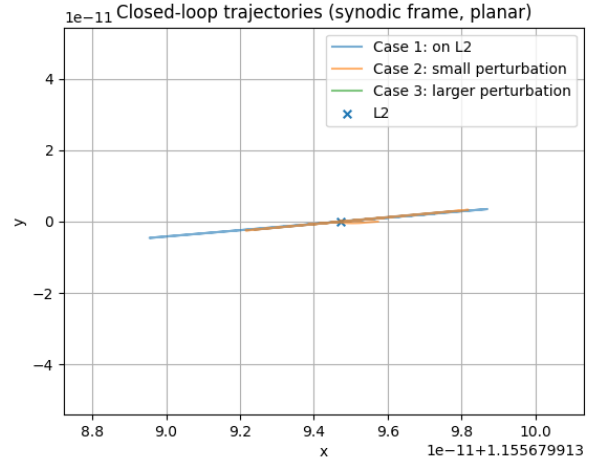


Fig. 4. Zoomed-in version of Fig. 3. Case 3 terminated before near the top of the figure before it fully converged to L_2 . Cases 1 and 2 both slightly oscillate along one of the eigenvectors while staying near L_2 .

Fig. 5 further demonstrates the exponential convergence for case 3. The oscillations during its convergence can be attributed to the non-zero imaginary components of the system's eigenvalues. Cases 1 and 2 also show some degree of oscillation, but given the amplitude of that, it may simply be the result of numerical errors and machine precision.

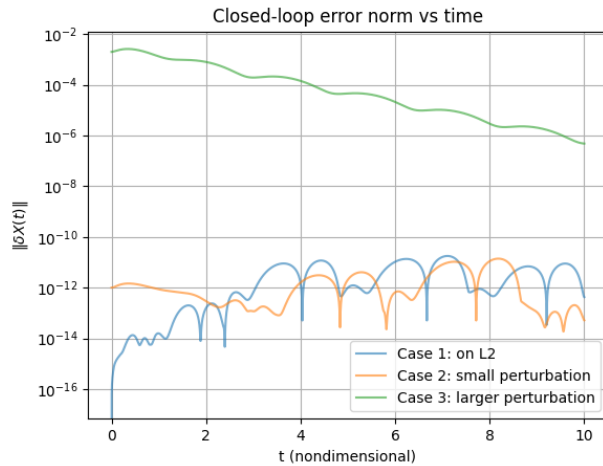


Fig. 5. $X - X^*$ error with the vertical axis displayed in log scale.

VII. DISCUSSION AND CONCLUSION

The analysis and simulations in the above sections have shown that the second Lagrange point in a Circular Restricted Three Body Problem (CRTBP) is unstable, due to the existence of an eigenvalue with positive real part. While there exists a family of periodic orbits known as Lyapunov orbits, spacecraft operating near this location should employ closed loop controllers for station-keeping. Assuming perfect state estimation, it was further shown that these closed loop controllers may be as simple as a Linear Quadratic Regulator. Analysis using eigenvalues and Lyapunov direct method then rigorously showed that with even a simple linear continuous time control strategy, the spacecraft's position can successfully converge to L_2 , from at least as far as 10^{-3} Earth-Moon distance (384km) apart. Granted, this result may vary with added restrictions, namely control effort caps imposed by thruster capabilities, fuel budget, or mission time requirements, but these results can, at minimum, confirm the feasibility of long term space missions near L_2 . Operators of an L_2 communication satellite, for example, may want to reduce R to save on fuel, and increase the position components of Q to ensure that the spacecraft remains within the designated section of space.

The scope of this paper covers only the dynamics of a spacecraft at L_2 , and only offered a cursory discussion about Lyapunov orbits. A more interesting topic to explore would be to maintain a spacecraft on one of the Lyapunov orbits, and potentially expand into a hybrid between planar and vertical Lyapunov orbits, such as the special case known as Halo orbits.

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